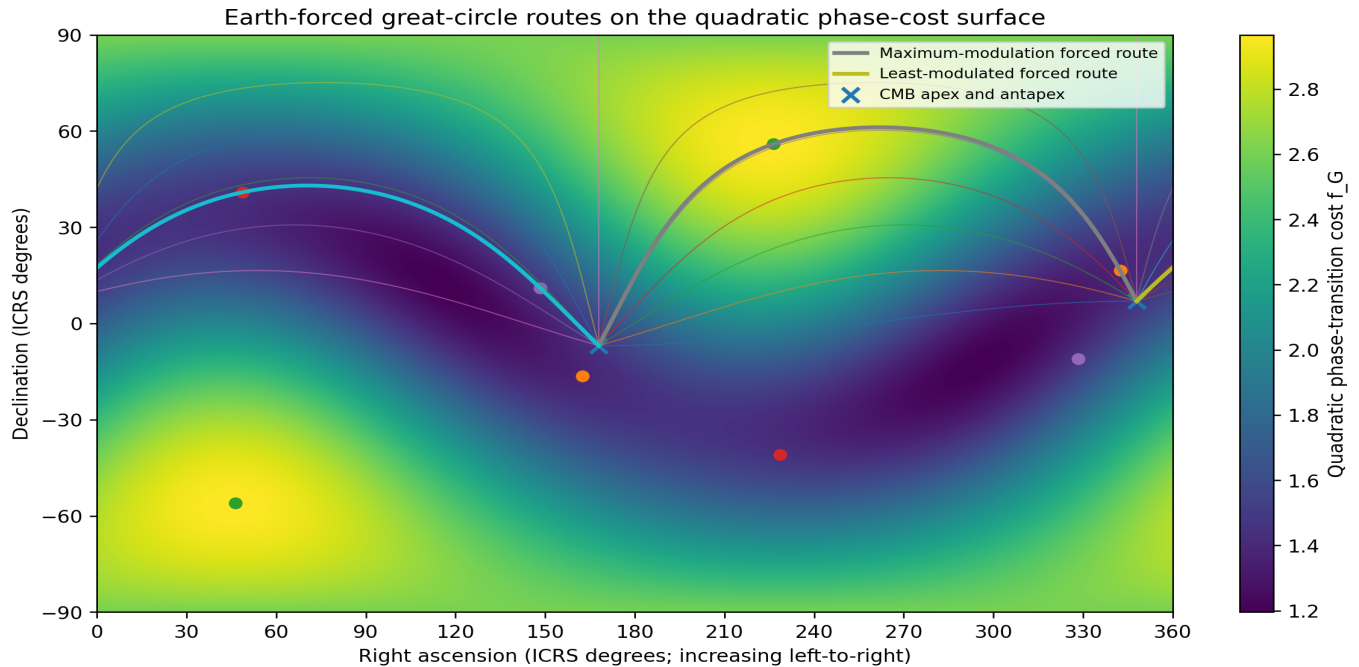


K-FIELD

Earth-CMB Modulation Routes

Exact kinematic forcing, unit-sphere great-circle trajectories, optimized phase-cost modulation, and native Cell3 orientation



Title figure. The authoritative rectangular ICRS phase-transition cost surface with a representative family of Earth/CMB-forced great-circle meridians. The optimized least-modulated and maximum-modulation routes, CMB apex and antapex, and route extrema are computed directly from the stated metric and vector.

Principal Investigator

P. Dwayne Esterline

BS, Huntington University (1999)

MBA, Indiana Wesleyan University (2008)

Krytronx

PROPRIETARY CONFIDENTIAL TRADE SECRET

Archival technical reference. Distribution and use are restricted. All numerical figures are generated from the actual metric, vectors, route solutions, and native Cell3 geometry stated in this document.

Publication epoch: 1784647098

Contents

Contents	2
Abstract	3
1. Definitions, assumptions, and coordinate lock	3
2. Authoritative quadratic phase-transition metric	3
3. Tangent projection of the Earth trajectory vector	4
3.1 Exact solution	4
3.2 Great-circle foliation	5
4. Phase-cost evolution along forced trajectories	5
5. Constrained optimization over Earth-forced route planes	6
6. Maximum-modulation Earth-forced route	7
7. Least-modulated Earth-forced route	8
8. Relationship to the two exact zero-modulation great circles	9
9. Strongest instantaneous forced ascent and descent	9
10. Route taxonomy and physical interpretation	9
11. Native Cell3 base cell and oriented planes	10
11.1 Complete centered half-step node table	12
12. Reproduction procedure	13
13. Canonical result statement	13
Archival notice	13

Abstract

This paper develops the complete Earth-centered directional kinematics obtained when the fixed Earth/CMB trajectory direction is projected onto the tangent plane of the authoritative ICRS unit sphere. The resulting differential equation is solved exactly. Every integral curve is shown to be a great-circle semicircle connecting the CMB antapex to the CMB apex for pull coupling, with reversed traversal for push coupling. The fixed quadratic phase metric then determines whether each local segment is uphill or downhill in transition cost.

The constrained family is optimized without changing coordinate systems. The maximum-modulation Earth-forced route has cost range 1.375987168544142 to 2.966117264626125, while the least-modulated forced route has cost range 1.449852380192369 to 1.467312133901670. The second exact zero-modulation great circle is only 0.676079472230 degrees from containing the CMB vector, explaining the existence of a nearly constant-cost forced route. The strongest instantaneous pull-driven ascent and descent are also identified.

The analysis is dimensionless on the unit sphere. The angular response scale Omega_E carries units of inverse time. Native Cell3 geometry is included as the underlying ICRS-oriented base cell, with all eight vertices, all 27 centered half-step nodes, primitive face areas, normals, and volume.

1. Definitions, assumptions, and coordinate lock

The directional state is a unit vector $\mathbf{n}$ in the fixed ICRS Cartesian basis. The x axis is RA 0 degrees and Dec 0 degrees, the y axis is RA 90 degrees and Dec 0 degrees, and the z axis is Dec +90 degrees. Right ascension increases from 0 to 360 degrees without inversion, wrapping, reflection, or conversion to another astronomical frame.

$$\mathbf{S}^2_{\text{ICRS}} = \{ \mathbf{n} \text{ in } \mathbb{R}^3 : \mathbf{n}^T \mathbf{n} = 1 \}$$
$$\mathbf{n}(\text{RA}, \text{Dec}) = \begin{bmatrix} \cos(\text{Dec}) \cos(\text{RA}), \\ \cos(\text{Dec}) \sin(\text{RA}), \\ \sin(\text{Dec}) \end{bmatrix}^T$$

The device is placed at the Earth center for the present model. This removes daily surface-position geometry. Seasonal modulation of the Earth velocity is neglected, so the Earth/CMB direction is constant in ICRS. The model addresses directional kinematics only; the response-rate coefficient Omega_E remains an explicit physical coupling parameter.

Quantity	Authoritative value	Role
Earth/CMB unit direction	(-0.970770855836, 0.207348009111, -0.120874929482)	Fixed forcing axis in ICRS
Earth/CMB apex	RA 167.943299879828 deg, Dec -6.942599922130 deg	Pull attractor
Earth/CMB antapex	RA 347.943299879828 deg, Dec +6.942599922130 deg	Pull source / push attractor
Device position	Earth center	No surface latitude or sidereal rotation term
Seasonal velocity term	Neglected	Constant forcing vector
Angular response scale	Omega_E [time^-1]	Sets speed but not route geometry

2. Authoritative quadratic phase-transition metric

The phase-transition cost is generated directly by the symmetric positive-definite matrix G_phase in the same ICRS Cartesian basis. No rotation to an eigenframe is applied when evaluating or plotting the field.

$$\mathbf{G}_{\text{phase}} = \begin{bmatrix} 2.9764106378 & 1.2431507795 & -2.0803600354 \\ 1.2431507795 & 2.5922286635 & -2.3178839937 \\ -2.0803600354 & -2.3178839937 & 6.7735709133 \end{bmatrix}$$
$$\begin{aligned} q(\mathbf{n}) &= \mathbf{n}^T \mathbf{G}_{\text{phase}} \mathbf{n} \\ f_{\text{G}}(\mathbf{n}) &= \sqrt{q(\mathbf{n})} \end{aligned}$$

The eigenvalues are 1.4293287534874834, 2.1125450264474788, and 8.800336434665038. They establish the unconstrained global soft, intermediate, and hard cost scales. The Earth-forced routes examined below are a constrained subset of all great circles

because their planes must contain the Earth/CMB axis.

Unit sphere, Earth/CMB axis, and optimized Earth-forced route planes

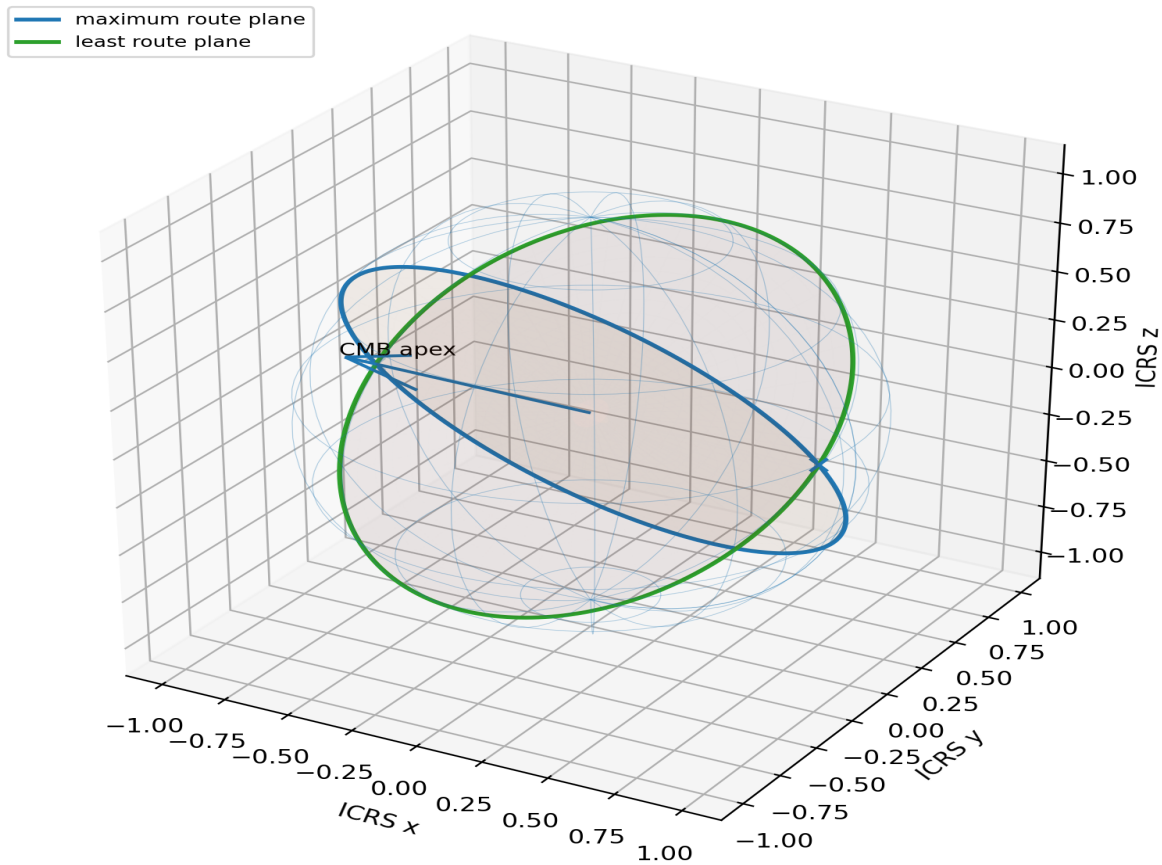


Figure 1. Three-dimensional unit sphere showing the fixed CMB axis and the two optimized Earth-forced route planes. Every valid forced plane contains the CMB vector; therefore its normal lies in the plane perpendicular to that vector.

3. Tangent projection of the Earth trajectory vector

A directional state constrained to the unit sphere cannot acquire a radial component. The admissible component of a constant Earth trajectory vector is its projection into the tangent plane at $\mathbf{n}$. The tangent projector is $\mathbf{P}_n = \mathbf{I} - \mathbf{n} \mathbf{n}^T$.

$$\mathbf{P}_n = \mathbf{I} - \mathbf{n} \mathbf{n}^T$$

$$\begin{aligned} \mathbf{V}_{\text{tan}} &= \mathbf{V}_E \mathbf{P}_n \mathbf{c}_{\text{hat}} \\ &= \mathbf{V}_E [\mathbf{c}_{\text{hat}} - (\mathbf{c}_{\text{hat}} \cdot \mathbf{n}) \mathbf{n}] \end{aligned}$$

The minimum-structure directional response law replaces the dimensional speed factor by the angular response rate Ω_E and includes a coupling sign σ .

$$\frac{d\mathbf{n}}{dt} = \sigma \Omega_E [\mathbf{c}_{\text{hat}} - (\mathbf{c}_{\text{hat}} \cdot \mathbf{n}) \mathbf{n}]$$

$$\begin{aligned} \sigma &= +1 : \text{pull toward the CMB apex} \\ \sigma &= -1 : \text{push toward the CMB antapex} \end{aligned}$$

The direction of the tangent vector is fixed by geometry. Ω_E controls only traversal rate. The tangent speed is zero at the apex and antapex, where the projected vector vanishes, and reaches its maximum on the great circle orthogonal to the CMB axis.

$$|d\mathbf{n}/dt| = \Omega_E \sqrt{1 - (\mathbf{c}_{\text{hat}} \cdot \mathbf{n})^2}$$

3.1 Exact solution

Choose any unit transverse vector $\mathbf{b}$ satisfying $\mathbf{b} \cdot \mathbf{c}_{\text{hat}} = 0$. The exact pull solution passing through $\mathbf{b}$ at $t = 0$ is:

$$n_b^{(+)}(t) = \tanh(\Omega_E t) \hat{c} + \operatorname{sech}(\Omega_E t) b$$

The corresponding push trajectory reverses the CMB-axis component:

$$n_b^{(-)}(t) = -\tanh(\Omega_E t) \hat{c} + \operatorname{sech}(\Omega_E t) b$$

The identity $\tanh^2(x) + \operatorname{sech}^2(x) = 1$ proves that the trajectory remains on the unit sphere. Direct differentiation reproduces the tangent-projection differential equation. For pull coupling, the remote past approaches the antapex and the remote future approaches the apex.

$$\lim_{t \rightarrow -\infty} n(t) = -\hat{c}$$

$$\lim_{t \rightarrow +\infty} n(t) = +\hat{c}$$

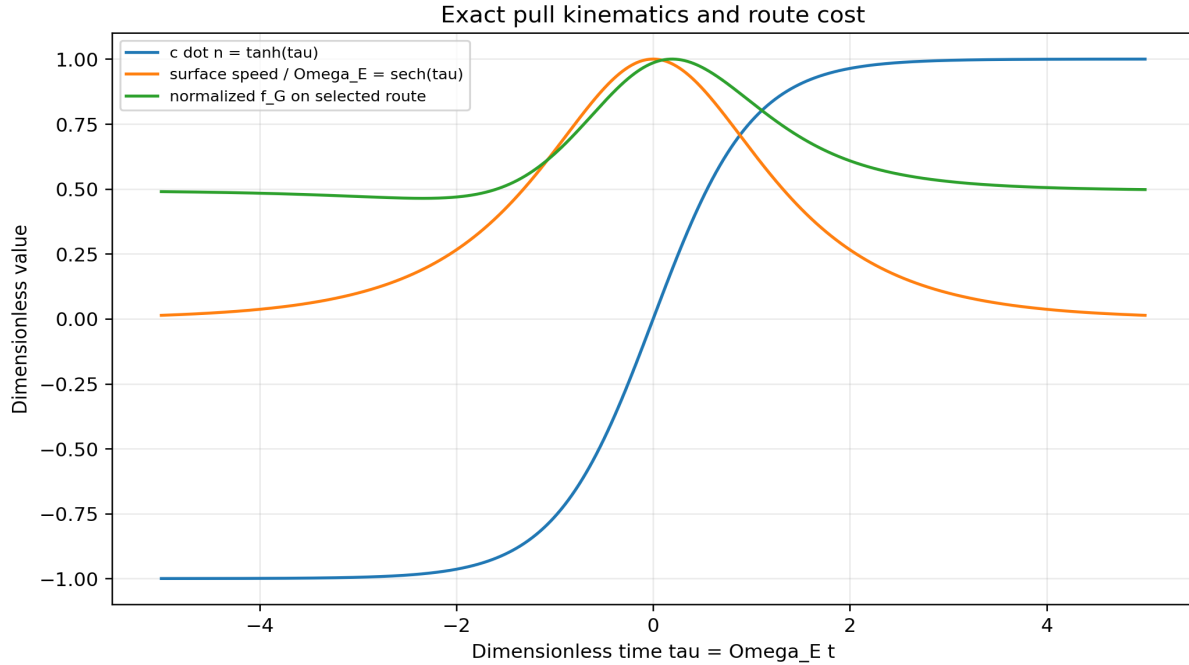


Figure 2. Exact dimensionless pull kinematics. The CMB projection $\hat{c} \cdot n = \tanh(\tau)$, the normalized surface speed is $\operatorname{sech}(\tau)$, and the route cost follows the selected maximum-modulation semicircle. Here $\tau = \Omega_E t$.

3.2 Great-circle foliation

Each exact solution is confined to the two-dimensional plane spanned by $\hat{c}$ and b . Because that plane passes through the origin, its intersection with the unit sphere is a great circle. The plane normal is $m = \hat{c} \times b$, and every Earth-forced great circle therefore satisfies $m \cdot \hat{c} = 0$.

$$m = \hat{c} \times b$$

$$m \cdot \hat{c} = 0$$

The Earth trajectory vector does not select a single route. It produces a complete meridional foliation: all great-circle semicircles connecting antapex to apex. Opposite choices b and $-b$ select the two complementary semicircles of the same great circle.

4. Phase-cost evolution along forced trajectories

Differentiating the quadratic cost along the forced motion gives the exact local cost rate. This step combines the kinematic forcing field with the fixed ICRS phase metric without introducing a guessed path.

$$\frac{dq}{dt} = 2 \sum \Omega_E \left[\hat{c}^T G_{\text{phase}} n - (\hat{c} \cdot n) q(n) \right]$$

$$\frac{df_G}{dt} = \sum \Omega_E / f_G$$

$$* \left[\hat{c}^T G_{\text{phase}} n \right]$$

$$- (\hat{c} \cdot n) q(n)]$$

Define $H(n)$ as the bracketed scalar. Under pull coupling, $H > 0$ is uphill motion, $H < 0$ is downhill motion, and $H = 0$ identifies a turning point of the transition cost along the forced route. Push coupling reverses these classifications.

$$H(n) = \hat{c}^T G_{\text{phase}} n - (\hat{c} \cdot n) q(n)$$

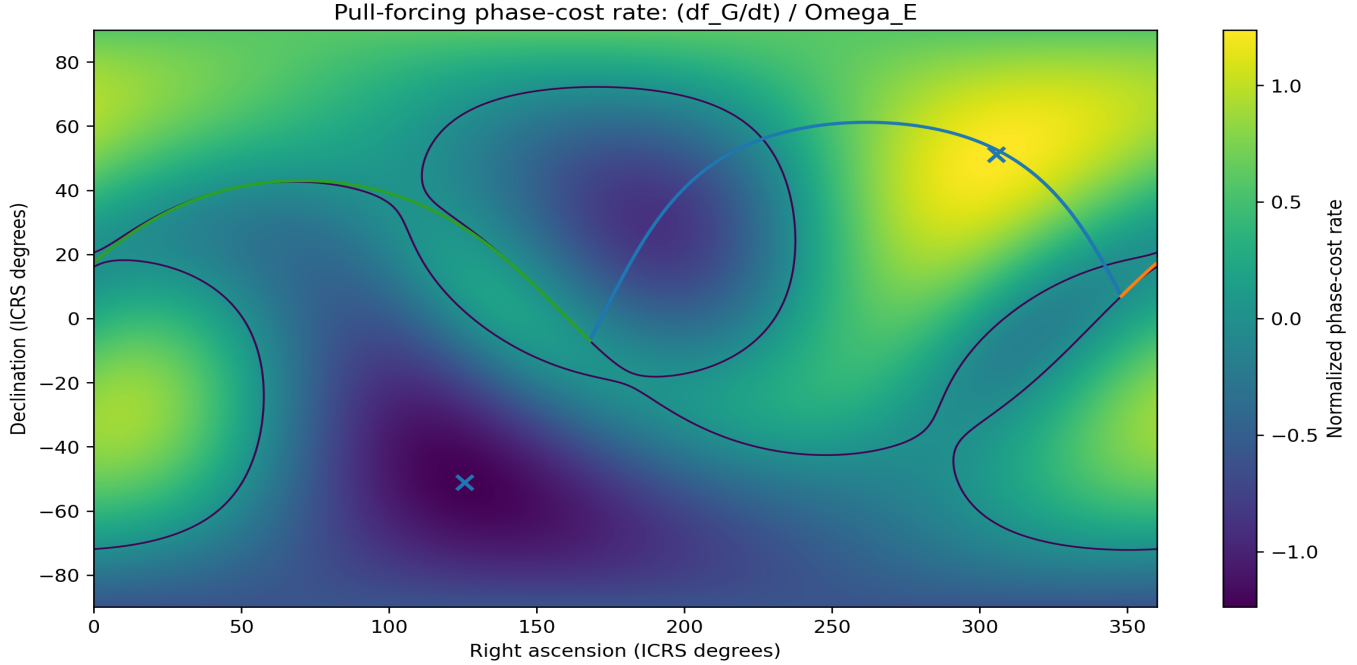


Figure 3. Normalized pull-forcing cost rate $(df_G/dt)/\Omega_E$ over the complete ICRS sphere. The zero contour separates local uphill and downhill sectors. The optimized forced routes and the strongest instantaneous ascent/descent points are overlaid.

Direct evaluation at the CMB fixed directions gives $q = 2.142872035667380$ and $f_G = 1.463855196277070$. Antipodal symmetry makes the pull starting and ending costs equal even though an individual route may encounter much lower and much higher intermediate costs.

5. Constrained optimization over Earth-forced route planes

Every allowed plane normal lies in the two-dimensional space perpendicular to $\hat{c}$. Choose an orthonormal basis e_A, e_B of that space and parameterize the projective plane-normal family by one angle ϕ :

$$m(\phi) = \cos(\phi)e_A + \sin(\phi)e_B, \\ 0 \leq \phi < \pi$$

For each normal m , the route plane has basis $B = [\hat{c}, b]$, where b is proportional to $m \times \hat{c}$. The restriction of G_{phase} to the route plane is the two-by-two matrix $R = B^T G_{\text{phase}} B$. Its eigenvalues μ_{minus} and μ_{plus} are the minimum and maximum q values attained on the full great circle.

$$R(m) = B^T G_{\text{phase}} B$$

$$\Delta f_G(m) = \sqrt{\mu_{\text{plus}}(m)} - \sqrt{\mu_{\text{minus}}(m)}$$

The complete one-parameter family was evaluated and locally refined. The maximum and minimum are isolated and reproduce the canonical route normals to projective angular precision substantially below the reporting resolution.

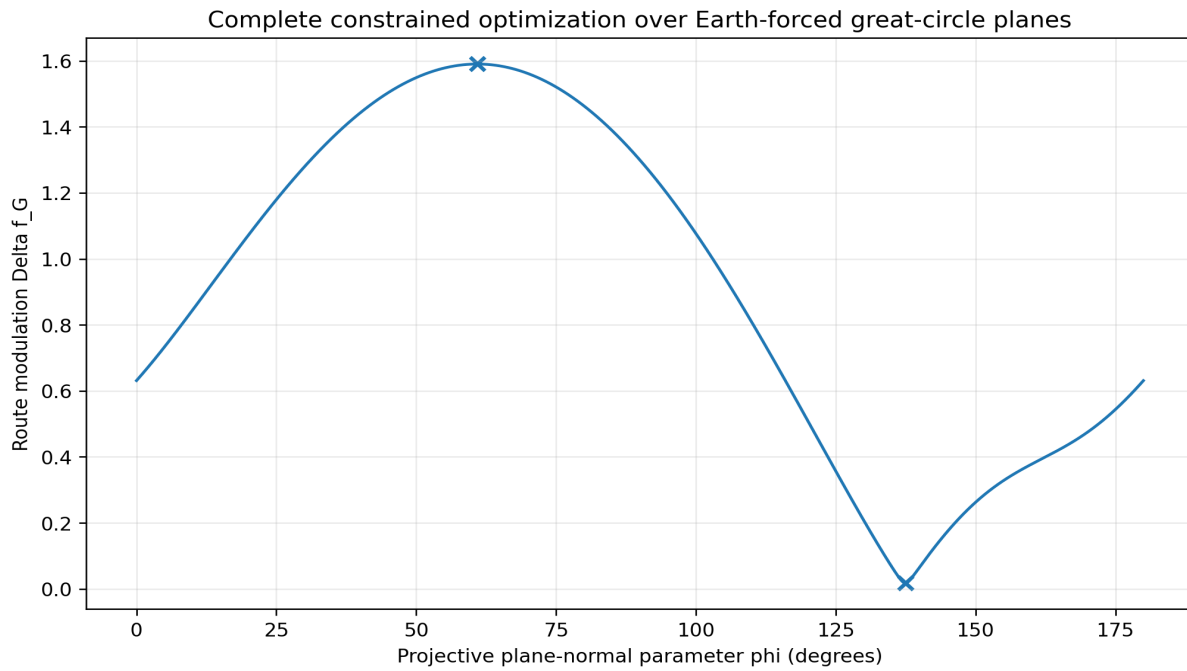


Figure 4. Transition-cost modulation over the complete projective family of Earth-forced great-circle planes. The marked points are the maximum-modulation and least-modulated constrained routes.

6. Maximum-modulation Earth-forced route

The greatest transition-cost range among all planes constrained to contain the CMB vector is obtained with plane normal:

```
m_max = (0.125128223569,
          0.867007487586,
          0.482328668167)
```

```
RA = 81.787665048092 deg
Dec = +28.837601767111 deg
```

The antipodal normal at RA 261.787665048092 degrees and Dec -28.837601767111 degrees represents the same plane. Its equation is:

```
0.125128223569 x
+ 0.867007487586 y
+ 0.482328668167 z = 0
```

The restricted eigenvalues are $\mu_{\text{minus}} = 1.893340687998125$ and $\mu_{\text{plus}} = 8.797851627513165$. The corresponding phase-cost range is:

```
f_min = 1.375987168544142
f_max = 2.966117264626125
Delta f_G = 1.590130096081983
```

Route feature	Primary position	Antipodal position	q	f_G
Low-cost point	RA 342.415652540024 deg; Dec +16.475701840803 deg	RA 162.415652540024 deg; Dec -16.475701840803 deg	1.893340687998133	1.375987168544145
High-cost point	RA 226.433871799486 deg; Dec +55.976636769738 deg	RA 46.433871799486 deg; Dec -55.976636769738 deg	8.797851627513181	2.966117264626127

One pull branch travels from the CMB antapex through the low-cost point, then climbs through the high-cost point, and finally descends to the CMB apex. The complementary branch encounters the antipodal high- and low-cost points in the opposite spherical half-plane.

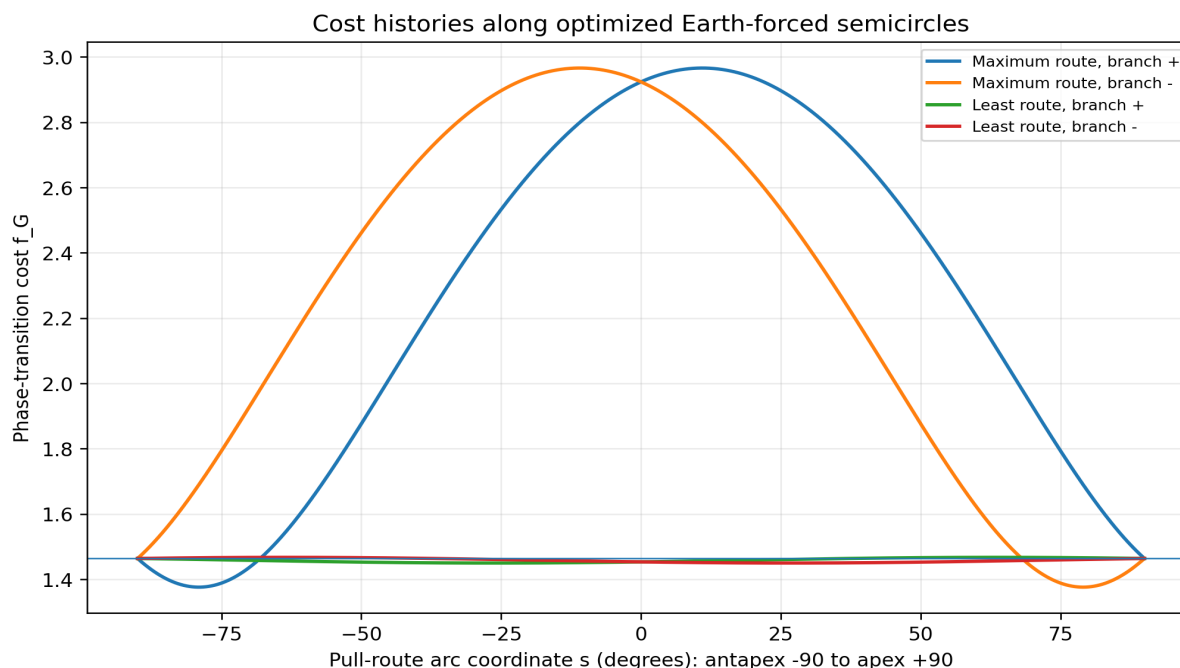


Figure 5. Complete phase-cost histories for both semicircular branches of the maximum and least forced routes. Every curve begins and ends at the common apex/antapex cost, but their intermediate histories differ strongly.

7. Least-modulated Earth-forced route

The smallest cost modulation available under the requirement that the route plane contain the Earth/CMB axis is obtained with:

```
m_min = (0.228327642017,
          0.642651833472,
          -0.731344726395)
```

```
RA = 70.440393013106 deg
Dec = -46.999245982318 deg
```

Its antipodal normal is RA 250.440393013106 degrees and Dec +46.999245982318 degrees. The plane equation is:

```
0.228327642017 x
+ 0.642651833472 y
- 0.731344726395 z = 0
```

The restricted values are $\mu_{\text{minus}} = 2.102071924349477$ and $\mu_{\text{plus}} = 2.153004898295072$, giving:

```
f_min = 1.449852380192369
f_max = 1.467312133901670
Delta f_G = 0.017459753709301
```

Route feature	Primary position	Antipodal position	q	f_G
Low-cost point	RA 48.671566552606 deg; Dec +40.893812972710 deg	RA 228.671566552606 deg; Dec -40.893812972710 deg	2.102071924349914	1.449852380192519
High-cost point	RA 148.383572602258 deg; Dec +11.022639371845 deg	RA 328.383572602258 deg; Dec -11.022639371845 deg	2.153004898295515	1.467312133901821

This route is nearly constant-cost. Its modulation is about one percent of the maximum constrained modulation. It is therefore the natural Earth-forced approximation to a zero-modulation trajectory.

8. Relationship to the two exact zero-modulation great circles

An exact zero-modulation great circle is Earth-forced only if its plane contains $c_{\hat{}}$. Equivalently, its plane normal must be perpendicular to $c_{\hat{}}$. The first exact zero-modulation plane misses the CMB vector by 20.407121644366 degrees. The second misses it by only 0.676079472230 degrees.

Reference plane	Plane-normal RA	Plane-normal Dec	CMB distance from plane
Exact zero-modulation route 1	191.747587256914 deg	+60.050129760577 deg	20.407121644366 deg
Exact zero-modulation route 2	69.468568665584 deg	-46.869771113084 deg	0.676079472230 deg
Least-modulated Earth-forced route	70.440393013106 deg	-46.999245982318 deg	0 by construction

The projective angular separation between the least-forced plane normal and the second zero-modulation plane normal is 0.676102845886 degrees. The constrained solution is therefore a small rotation of the second zero-modulation plane. This geometric near-alignment explains why the forced minimum modulation is small but not exactly zero.

9. Strongest instantaneous forced ascent and descent

Maximizing the exact normalized derivative $(df_G/dt)/\Omega_E$ over the full sphere gives the position of greatest instantaneous pull-driven ascent:

RA = 305.538311619787 deg
Dec = +51.174520573108 deg

$(df_G/dt)/\Omega_E = +1.237871772310$

The antipodal position is the strongest instantaneous descent:

RA = 125.538311619787 deg
Dec = -51.174520573108 deg

$(df_G/dt)/\Omega_E = -1.237871772310$

These points lie on an Earth-forced great circle whose plane normal is RA 82.018041820498 degrees and Dec +30.265751931271 degrees. This route is close to, but distinct from, the maximum-total-modulation route. The distinction is exact: one optimization maximizes the local slope, while the other maximizes the full peak-to-peak cost range.

10. Route taxonomy and physical interpretation

The Earth/CMB vector creates a continuous family of admissible directional trajectories. Every pull route begins at the antapex and terminates at the apex. Every push route reverses the same geometry. The phase metric partitions each route into uphill, downhill, and turning-point segments.

The least-modulated route provides an approximately neutral corridor. The maximum-modulation route provides the largest available loading and unloading range while remaining dynamically aligned with the Earth forcing field. Intermediate plane normals generate a continuum of routes with intermediate modulation. Route selection therefore changes the cost history without changing the fixed Earth/CMB endpoints.

The static directional model does not assign a physical energy unit to f_G . A device model must specify how a physical state couples to n , how Ω_E depends on Earth motion and device parameters, and how phase-cost changes map to mechanical, electromagnetic, or other work channels. Within the present kinematics, the exact route structure required for such modeling is fully determined.

Earth-forced family:
- $c_{\hat{}}$ ---> + $c_{\hat{}}$ for pull coupling
+ $c_{\hat{}}$ ---> - $c_{\hat{}}$ for push coupling

Available constrained modulation:
least route $\Delta f_G = 0.017459753709301$
maximum route $\Delta f_G = 1.590130096081983$

11. Native Cell3 base cell and oriented planes

The route dynamics are defined on the ICRS unit sphere. The K-Field base geometry is recorded independently in native unit-sphere-derived lengths. The ordered primitive translations a , b , and c are the columns of the native Cell3 basis. They are dimensionless native lengths rather than nanometres or megaparsecs.

$$a = \begin{pmatrix} 0.0203814385169464492, \\ -0.00341641449641054938, \\ -0.00233296591672896075 \end{pmatrix}$$

$$b = \begin{pmatrix} 0.00219874596832335531, \\ 0.0129021702586925406, \\ -0.0340665015360557127 \end{pmatrix}$$

$$c = \begin{pmatrix} -0.000630379485347372023, \\ -0.0242577220171278675, \\ -0.042924354571093816 \end{pmatrix}$$

Authoritative native Cell3 base cell: 8 vertices and 27 centered half-step nodes

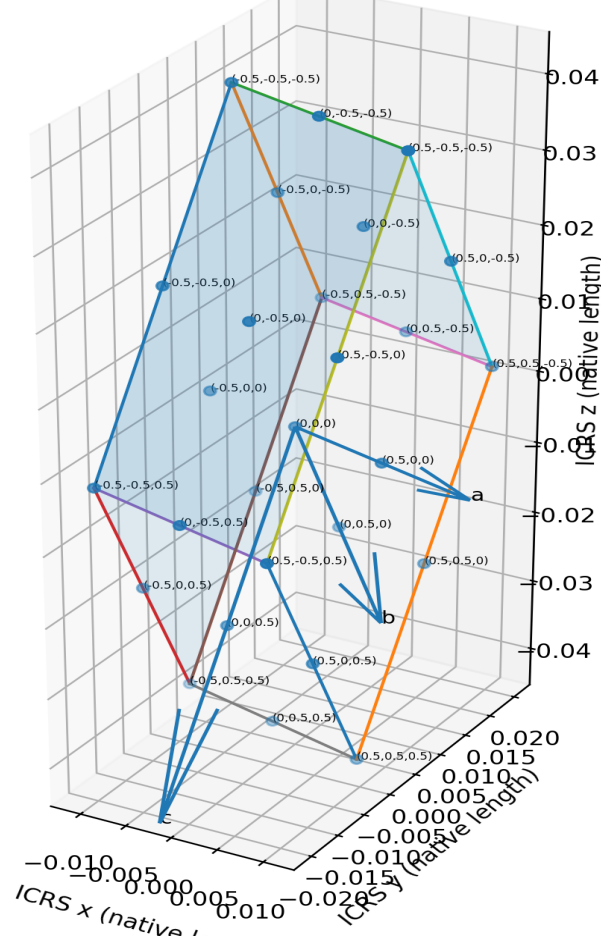


Figure 6. Dimensionally correct centered Cell3 in the authoritative ICRS orientation. The eight vertices are the sign combinations of one-half a , b , and c . All 27 centered half-step nodes are shown and labeled by their coefficient triples in $\{-1/2, 0, +1/2\}^3$.

Cell3 primitive faces, eight vertices, and oriented face normals

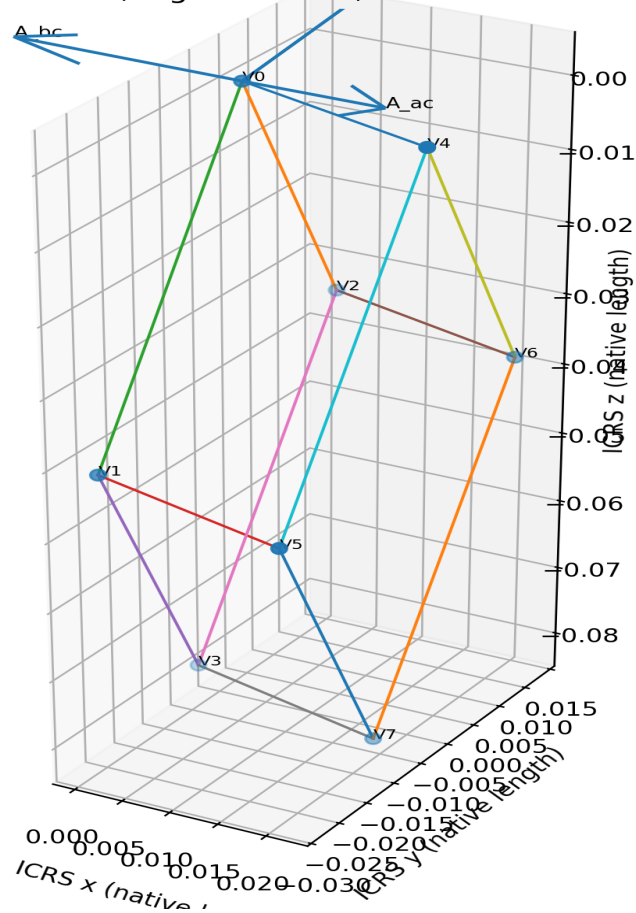


Figure 7. Primitive Cell3 rendering with all eight vertices and the three oriented face normals. The ab, ac, and bc planes retain the exact native geometry and ICRS orientation.

Vector	ICRS x	ICRS y	ICRS z	Native length
a	0.0203814385169464492	-0.00341641449641054938	-0.00233296591672896075	0.0207970587824329485
b	0.00219874596832335531	0.0129021702586925406	-0.0340665015360557127	0.0364942051307271878
c	-0.000630379485347372023	-0.0242577220171278675	-0.042924354571093816	0.0493085658998594217

Face	Area [native^2]	Included angle	Unit normal (ICRS)
A_ab	0.00075472178967262403	83.933483012313 deg	(0.194092200808, 0.913177168821, 0.358379234791)
A_ac	0.00101125575768806525	80.448130018338 deg	(0.089052594102, 0.866576769291, -0.491033948326)
A_bc	0.00138578443173328707	50.363231813756 deg	(-0.995965190309, 0.083602162507, -0.032619290564)

The primitive volume is $|\det[a \ b \ c]| = 2.84206693780964857e-05 \text{ native}^3$. The ordered basis is left-handed because $\det[a \ b \ c]$ is negative; distances, face areas, route geometry, and the symmetric phase metric remain well-defined.

11.1 Complete centered half-step node table

Node	Coefficient triple	ICRS x	ICRS y	ICRS z
N00	(-0.5, -0.5, -0.5)	-0.0109749024999612	0.00738598312742294	0.0396619110119392
N01	(-0.5, -0.5, 0)	-0.0112900922426349	-0.004742877881141	0.0181997337263923
N02	(-0.5, -0.5, 0.5)	-0.0116052819853086	-0.0168717388897049	-0.00326244355915457
N03	(-0.5, 0, -0.5)	-0.00987552951579954	0.0138370682567692	0.0226286602439114
N04	(-0.5, 0, 0)	-0.0101907192584732	0.00170820724820527	0.00116648295836448
N05	(-0.5, 0, 0.5)	-0.0105059090011469	-0.0104206537603587	-0.0202956943271824
N06	(-0.5, 0.5, -0.5)	-0.00877615653163786	0.0202881533861155	0.00559540947588353
N07	(-0.5, 0.5, 0)	-0.00909134627431155	0.00815929237755154	-0.0158667678096634
N08	(-0.5, 0.5, 0.5)	-0.00940653601698523	-0.00396956863101239	-0.0373289450952103
N09	(0, -0.5, -0.5)	-0.000784183241487992	0.00567777587921766	0.0384954280535748
N10	(0, -0.5, 0)	-0.00109937298416168	-0.00645108512934627	0.0170332507680279
N11	(0, -0.5, 0.5)	-0.00141456272683536	-0.0185799461379102	-0.00442892651751905
N12	(0, 0, -0.5)	0.000315189742673686	0.0121288610085639	0.0214621772855469
N13	(0, 0, 0)	0	0	0
N14	(0, 0, 0.5)	-0.000315189742673686	-0.0121288610085639	-0.0214621772855469
N15	(0, 0.5, -0.5)	0.00141456272683536	0.0185799461379102	0.00442892651751905
N16	(0, 0.5, 0)	0.00109937298416168	0.00645108512934627	-0.0170332507680279
N17	(0, 0.5, 0.5)	0.000784183241487992	-0.00567777587921766	-0.0384954280535748
N18	(0.5, -0.5, -0.5)	0.00940653601698523	0.00396956863101239	0.0373289450952103
N19	(0.5, -0.5, 0)	0.00909134627431155	-0.00815929237755154	0.0158667678096634
N20	(0.5, -0.5, 0.5)	0.00877615653163786	-0.0202881533861155	-0.00559540947588353
N21	(0.5, 0, -0.5)	0.0105059090011469	0.0104206537603587	0.0202956943271824
N22	(0.5, 0, 0)	0.0101907192584732	-0.00170820724820527	-0.00116648295836448
N23	(0.5, 0, 0.5)	0.00987552951579954	-0.0138370682567692	-0.0226286602439114
N24	(0.5, 0.5, -0.5)	0.0116052819853086	0.0168717388897049	0.00326244355915457
N25	(0.5, 0.5, 0)	0.0112900922426349	0.004742877881141	-0.0181997337263923
N26	(0.5, 0.5, 0.5)	0.0109749024999612	-0.00738598312742294	-0.0396619110119392

12. Reproduction procedure

A complete reproduction requires only the matrix G_{phase} , the normalized Earth/CMB vector, the ICRS conversion formula, and the native Cell3 basis. The following deterministic sequence reproduces every reported route and figure:

1. Normalize c_{hat} .
2. Form $P_n = I - n n^T$.
3. Integrate or evaluate the exact solution $n(t) = \tanh(\Omega_E t) c_{\text{hat}} + \text{sech}(\Omega_E t) b$.
4. Restrict each candidate plane with $B = [c_{\text{hat}}, b]$.
5. Compute $\text{eig}(B^T G_{\text{phase}} B)$.
6. Optimize Δf_G over $m \cdot c_{\text{hat}} = 0$.
7. Evaluate the exact rate field $H(n)/f_G(n)$.
8. Convert vectors to RA/Dec only for final presentation.

The companion JSON file records all inputs, matrices, route samples, extrema, Cell3 vertices and nodes, numerical validation errors, and figure names. It is sufficient to reproduce the full numerical content without extracting values from the PDF.

13. Canonical result statement

Under the explicit tangent-projection model, the fixed Earth/CMB trajectory vector generates a continuous family of great-circle routes joining the CMB antapex and apex. The route geometry is determined solely by the CMB direction. The authoritative phase metric determines the cost history on each route.

The family contains a nearly constant-cost route with $\Delta f_G = 0.017459753709301$ and a maximum-modulation route with $\Delta f_G = 1.590130096081983$, together with every intermediate modulation. The second exact zero-modulation plane is nearly aligned with the Earth-forcing foliation, requiring only a 0.676102845886-degree projective rotation to become an admissible Earth-forced plane.

The Earth vector does not globally force all directions uphill or all directions downhill. It forces motion along CMB meridians. The fixed ICRS quadratic metric divides those meridians into alternating uphill and downhill sectors. This is the explicit kinematic structure needed to construct and test Earth-driven directional route cycles.

Archival notice

Principal Investigator: P. Dwayne Esterline. This document and the contained K-Field formulations, numerical kernels, geometric constructions, analytical pathways, and emitted data products are designated PROPRIETARY CONFIDENTIAL TRADE SECRET material of Krytronx. Distribution and use are restricted. All Rights Reserved. Do Not Distribute.